



Assessment: AI/ML Data Annotation Intern

Hello,

Greetings of the day!

Space-based and ground-based sensors capture images and track "space objects," i.e., satellites/space debris. These objects & stars are visible as streaks/lines and spots/points against the dark/noisy background, respectively.

As an **AI/ML Data Annotation Intern**, you are tasked with annotating features such as blobs and streaks in the provided SSA imagery using an appropriate technique. You are required to pre-process and tile the imagery into the specified dimensions before creating accurate pixel-level masks to delineate the features. Consider the project described above; please answer the following.

Question 1:

Accurately annotate the features such as **stars/blobs** and **objects/streaks** in the real-sky images provided in an OneDrive folder using an appropriate technique.

- A1: Inspect the provided imagery and apply appropriate pre-processing steps, if required.
- A2: Divide each processed image into tiles with a target size of 1024×1024 pixels.
- A3: Annotate the blobs and streaks as two different classes by creating pixel-level masks to accurately delineate the blobs and streaks, respectively.

The typical parameters of the real-sky images given are as follows:

- Image dimension: 9568 x 6380 pixels
- Data format: FITS
- Data type: 16-bit unsigned integer
- Band: Single-band (grayscale)

Note: The dataset consists of 10 raw images stored in the Datasets_Assessment folder, which must be used for data annotation. A reference image and the respective annotation result are provided only for visual inspection of streaks and blobs. Please use the OneDrive link mentioned below to access the datasets.



Floor 9, Brigade Senate 2,
Hebbal, Bengaluru, India – 560024
info@digantara.co.in | www.digantara.co.in

Question 2

The annotation method used in Question 1 needs to be capable of producing an accurate pixel-level mask for the feature of interest. Please answer the following questions with clear reasoning based on your practical implementation.

- a) Detail the specific pre-processing steps you applied to the provided images.
- b) Explain your programmatic approach for handling the image boundaries when dividing the 9568 x 6380 images into exact 1024 x 1024 tiles without losing data.
- c) State the method and criteria for distinguishing between short/fat streaks and blobs while performing annotation.
- d) Specify the criteria established for annotating faint or small blobs and labelling them as the star class.

Note:

- Send the response as a zip file with clear instructions on how to access the contents. Export and submit the repatched full-size imagery for visual inspection, along with the corresponding annotations in YOLO Ultralytics segmentation format.
- If using custom scripts or AI/ML annotation pipelines or algorithms, share them through a GitHub/Git repository. If using UI-based annotation tools (e.g., CVAT, Roboflow), include a short screen recording (video) demonstrating your annotation workflow, along with the exported annotation result files.
- State assumptions in the relevant sections.
- Avoid generic answers.
- Provide the written response within four pages, excluding the source code and support files.

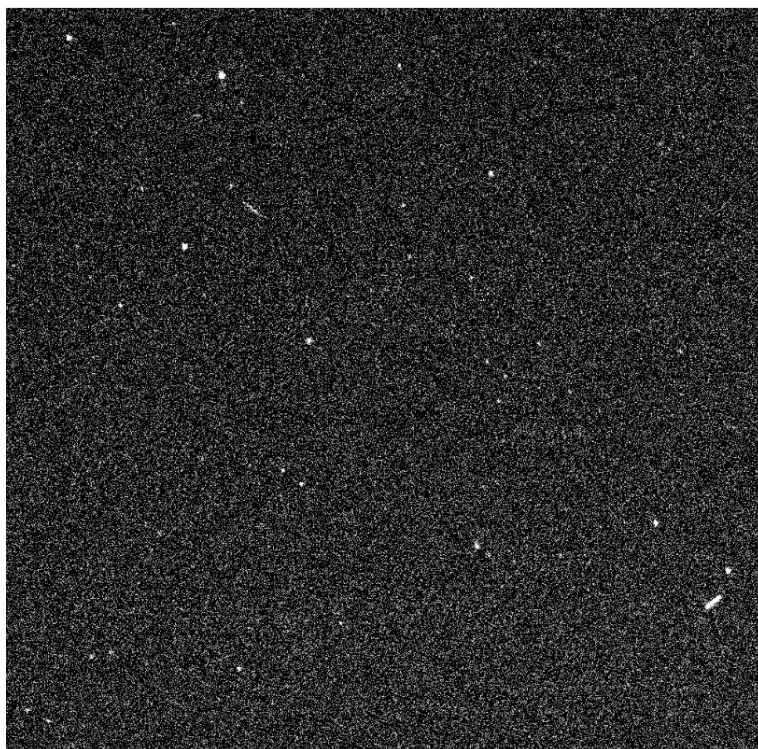


Fig. 1: Sample tile of raw image (1024 x 1024 pixels)

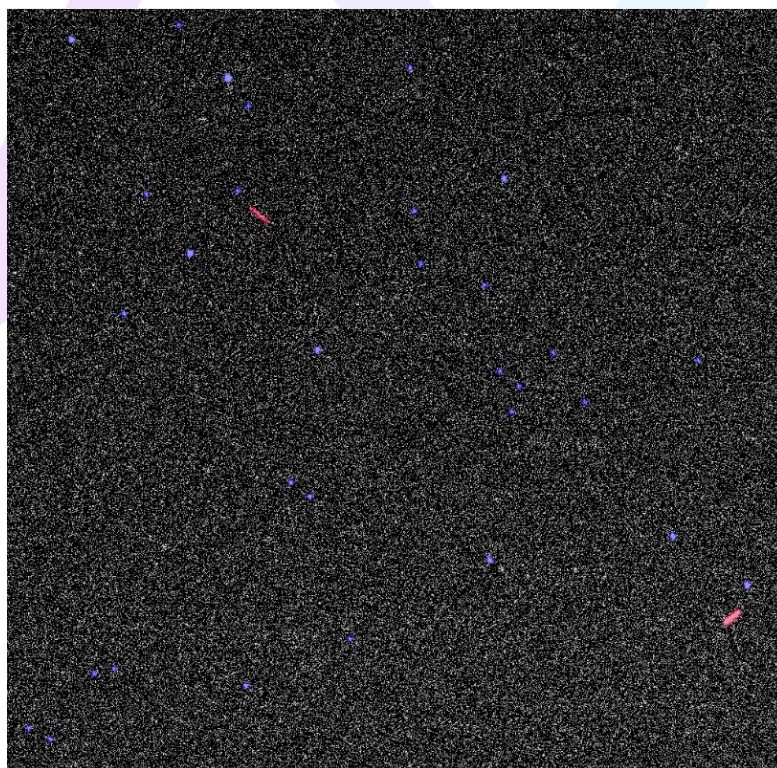


Fig. 2: Annotation result (pixel-level masks) for blobs and streaks